

Missouri University of Science and Technology

Intro to Data Science
CS 3402 (Spring 2026)

Lecture 19 Project Tutoring & Quiz 3 Material Review

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Today's Topics

1. Project 4 Tutoring

2. Quiz 3 Material Review

Group Project Topic 4 - LLMs

Title: Evaluating Prompt Engineering vs Fine-Tuning for Domain Adaptation in Large Language Models

- **Prompt Engineering:** Designing specific prompts (zero-shot or few-shot) without changing model weights.
- **Fine-Tuning:** Updating model weights on a task-specific dataset to adapt to the domain.

Which strategy is more effective for LLMs?

Group Project Topic 4 - Evaluating Prompt Engineering vs Fine-Tuning for Domain Adaptation in Large Language Models

1. Data Collection and Preprocessing

Dataset Requirements: At least 3 different datasets;
Task types: factual QA, reasoning, explanation, sentiment, etc.
Source of prompts: public datasets (e.g., MedQA, FinQA), translated or self-constructed prompts

Data inspection and quality check: Remove ambiguous or overly general prompts; Preprocess to standardize format and instruction phrasing
Dataset statistics: Number of tasks; Number of prompts per task type

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1. Data Collection and Preprocessing

Task Type	Dataset
Factual QA	TyDi QA
Reasoning	GSM8K
Causal Reasoning	XCOPA
Knowledge QA	MKQA

Dataset description; Data inspection and quality check; Dataset statistics; etc.

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2. Model Selection and Experimental Design

(a) Large Language Models (LLMs) - Closed-source models
Examples: **ChatGPT, Claude, Gemini,** or similar models
Zero-shot learning: no task-specific examples provided
Few-shot learning: a small number of labeled examples provided as prompts

Prompt design: Exact prompt templates used.

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2. Model Selection and Experimental Design

(b) Large Language Models (LLMs) - Open-source models

Examples: **Open-source LLMs** (e.g., LLaMA-based models, Falcon-40B, Mistral) or similar models
Learning settings. You should evaluate model performance under consistent and clearly defined settings:
Zero-shot learning: no task-specific examples provided
Few-shot learning: a small number of labeled examples provided as prompts

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2. Model Selection and Experimental Design

(c) Fine-Tuning Small LLMs

Fine-tune a smaller model (e.g., OpenAI GPT-2, Google T5) on a **subset of domain-specific prompts**

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3. Evaluation and Analysis

For all models:

- **Accuracy / Exact Match** – factual correctness
- **Reasoning Quality** – logical consistency, step-wise correctness
- **Response Characteristics** – length, completeness, hallucination rate
- **Optional Metrics** – BLEU/ROUGE for generation tasks, sentiment, or style

Analysis:

- Compare zero-shot, few-shot, and fine-tuned performance
- Assess cost vs performance: compute time, training requirements

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4. Documentation, Reporting, and Presentation

4.1 Project Report

- Introduction and motivation
- Dataset description and preprocessing steps
- Model descriptions
- Experimental setup
- Learning curve plots
- Tables comparing performance
- Key conclusions
- Limitations and future directions

Group Project Rubric (100 points total)

1. Data Collection and Preprocessing 10%
2. Model Selection and Experimental Design 30%
3. Evaluation and Analysis 20%
4. Project Report 20%
5. Project Presentation 20%

Quiz 3 Materials Review

Week 11	03/31: Quiz2 04/02: Language Models - Transformer ✅
Week 12	04/07: Vision Models – Vision Transformer ✅ 04/09: Vision Models – CNN & ResNet ✅ (Homework2 Out)
Week 13	04/14: Project Tutoring & Quiz 3 Material Review 04/16: Group Project Tutoring
Week 14	04/21: Group Project Presentation 04/23: Quiz 3

Quiz 3 Materials Review

True / False Questions

1. In CNNs, pooling layers reduce spatial dimensions.
2. Vision Transformers require convolution layers to process images.
3. Residual connections help mitigate vanishing gradient problems.
4. CNNs learn local spatial patterns using shared filters.
5. Self-attention allows each word to directly attend to all other words in a sentence.

Quiz 3 Materials Review

Multiple Choice

Which model is best suited for text generation?

A. Vision Transformer
B. GPT
C. CNN
D. ResNet

Quiz 3 Materials Review

What is the primary purpose of attention in Transformers?

A. Reduce image size

B. Learn word frequency

C. Model relationships between tokens

D. Perform pooling

Quiz 3 Materials Review

Which statement about CNNs is correct?

A. They use self-attention

B. They require positional encoding

C. They use shared convolution filters

D. They cannot process images

Quiz 3 Materials Review

In Vision Transformers, an image is first:

A. Downsampled using pooling

B. Split into patches

C. Converted into words

D. Passed through convolution layers

Quiz 3 Materials Review

Which component is NOT part of a Transformer encoder?

A. Multi-head attention

B. Feed-forward network

C. Recurrent layer

D. Positional encoding

Quiz 3 Materials Review

Tokenization

Explain the difference between:

Word-level tokenization

Subword tokenization

Also explain **why subword tokenization is preferred in modern LLMs**.

Quiz 3 Materials Review

List **two differences** between CNNs and Vision Transformers in image processing.

Quiz 3 Materials Review

A 224×224 image is split into 16×16 patches.

Tasks:

How many patches are created?

Why are patches flattened before entering Transformer?

Why do we need positional embeddings?

What does attention between patches represent intuitively?

Quiz 3 Materials Review

Self-Attention Calculation

Given a simple self-attention setup with **one token attending to two tokens**:

$$Q = [1, 0], \quad K_1 = [1, 1], \quad K_2 = [0, 1]$$

$$V_1 = [2, 0], \quad V_2 = [0, 3]$$

Tasks:

(a) Compute attention scores using dot product:

$$Q \cdot K_1, \quad Q \cdot K_2$$

(b) Apply softmax to obtain attention weights (you may leave answers in exponential form):

$$\text{softmax}(s_i) = \frac{e^{s_i}}{\sum e^{s_j}}$$

(c) Compute the final output vector:

$$\text{Output} = \alpha_1 V_1 + \alpha_2 V_2$$

Quiz 3 Materials Review

An image of size **32×32** is passed through a convolution layer with:

Filter size = 5×5; Stride = 1; Padding = 0

(a) Compute output width/height

(b) If we use **padding = 2**, what is the new output size?

(c) How many parameters per filter?

(d) Total parameters in the layer?

Quiz 3 Materials Review

Vision Transformer Patch Calculation

An image of size **256×256** is divided into patches of size **32×32**.

Tasks:

(a) Number of patches

(b) If each patch is flattened, what is the dimension of each patch vector (assume RGB image)?

(c) What is the total input sequence length to the Transformer?

CNN Forward Pass Calculation

You are given a grayscale input image:

$$X = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 4 & 1 & 2 & 1 \\ 0 & 1 & 3 & 2 \\ 2 & 2 & 1 & 0 \end{bmatrix}$$

A convolutional layer uses the following **kernel (filter)**:

$$K = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$$

Part A: Convolution (6 pts)

Perform convolution with:

- Stride = 1
- Padding = 0 (valid convolution)

Tasks:

(a) Compute the output feature map

(b) What is the output size?

Part B: Adding Padding (3 pts)

Now apply:

- Padding = 1
- Stride = 1

Tasks:

(c) What is the new output size? (No need to recompute full values)

Part C: Max Pooling (4 pts)

Take the output from **Part A** and apply:

- Max pooling
- Pool size = 2×2
- Stride = 2

Tasks:

(d) Compute the pooled output

(e) What is the output size after pooling?